Exercise 2

a) i) $\arctan'(x) = \dfrac{1}{\tan'(t)}$ where $t = \arctan(x)$

$$= \frac{1}{\left(\frac{\operatorname{sen}(t)}{\cos(t)}\right)'} = \frac{1}{\frac{\cos^2 t + \operatorname{sen}^2 t}{\cos^2 t}}$$

$$= \frac{1}{1+\tan^2(t)} = \frac{1}{1+x^2}$$

ii) $\arccos'(x) = \dfrac{1}{\cos'(t)}$ where $t = \arccos(t)$

$$= \frac{1}{-\sin(t)} = \frac{1}{-\sqrt{1-\cos^2 t}} = -\frac{1}{\sqrt{1-x^2}}$$

iii) $\sinh'(x) = \left(\dfrac{e^x - e^{-x}}{2}\right)' = \dfrac{e^x + e^{-x}}{2} = \cosh(x)$

b) i) $\displaystyle\int \frac{3}{25+x^2}\,dx =$

$$= \frac{3}{25}\int \frac{1}{1+\left(\frac{x}{5}\right)^2}\,dx \qquad \left(\text{substitution } \frac{x}{5} = t \quad dx = 5\,dt\right)$$

$$= \frac{3}{5}\int \frac{1}{1+t^2}\,dt \overset{\text{by part a)}}{=} \frac{3}{5}\arctan(t) + c = \frac{3}{5}\arctan\left(\frac{x}{5}\right) + c$$

ii) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx =$

$$= \frac{1}{3}\int \frac{1}{\sqrt{1-\left(\frac{x}{3}\right)^2}}\,dx = \qquad \left(\text{substitution } x/3 = t \quad dx = 3\,dt\right)$$

$$= \int \frac{1}{\sqrt{1-t^2}}\,dt = -\arccos(t) + c = -\arccos(x/3) + c$$

iii) $\displaystyle\int \sqrt{1+x^2}\,dx =$ (substitution $x = \sinh(t)$ $dx = \cosh(t)\,dt$)

$$= \int \sqrt{1+\sinh(t)^2}\,\cosh(t)\,dt = \int \cosh^2(t)\,dt$$

$$= \frac{1}{4}\int e^{2t} + 2 + e^{-2t}\,dt = \frac{1}{8}e^{2t} - \frac{1}{8}e^{-2t} + \frac{1}{2}t + c$$

$$= \frac{1}{2}\left(\frac{e^t + e^{-t}}{2}\right)\left(\frac{e^t - e^{-t}}{2}\right) + \frac{1}{2}t + c$$

$$= \frac{1}{2}\cosh(t)\operatorname{senh}(t) + \frac{1}{2}t + c$$

$$= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\operatorname{senh}(x) + c.$$

Exercise 5

a) We want to write

$$\frac{1}{(x+3)(2x-1)} = \frac{A}{x+3} + \frac{B}{2x-1}$$

$$= \frac{2Ax - A + Bx + 3B}{(x+3)(2x-1)}$$

We must impose $\begin{cases} 2A + B = 0 \\ -A + 3B = 1, \end{cases}$ yeldind $A = -\frac{1}{7}$, $B = \frac{2}{7}$

$$\int \frac{1}{2x^2+5x-3}\,dx = \int -\frac{1}{7}\frac{1}{x+3} + \frac{2}{7}\frac{1}{2x-1}\,dx$$

$$= -\frac{1}{7}\log(|x+3|) + \frac{1}{7}\log(|2x-1|) + c$$

b) Now we try to write

$$\frac{x}{(x-1)(x-2)^2} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$$= \frac{Ax^2 - 4Ax + \boxed{4A} + Bx^2 - 3Bx + \boxed{2B} + Cx + \boxed{C}}{(x-1)(x-2)^2}$$

Then we have to ask

$$\begin{cases} A + B = 0 \\ -4A - 3B + C = 1 \\ 4A + 2B - C = 0 \end{cases} \Rightarrow \begin{cases} A = 1 \\ B = -1 \\ C = 2 \end{cases}$$

$$\int \frac{x}{(x-1)(x-2)^2}\,dx = \int \frac{1}{x-1} - \frac{1}{x-2} + \frac{2}{(x-2)^2}\,dx$$

$$= \log(|x-1|) - \log(|x-2|) - \frac{2}{x-2} + C$$

Exercise 1

a) i) $\int x^3(x^4+1)\,dx =$ $\left(\text{substitution: } \begin{array}{l} x^4+1 = t \\ dt = 4x^3dx \end{array}\right)$

$= \int \frac{t^3}{4}\,dt$

$= \frac{t^4}{16} + c = \frac{(x^4+1)^4}{16} + c$

ii) $\int \frac{x}{\sqrt{x+1}}\,dx =$ $\left(\text{substitution: } \begin{array}{l} x+1 = t \\ dt = dx \end{array}\right)$

$= \int \frac{t-1}{\sqrt{t}}\,dt = \int \sqrt{t} - \frac{1}{t}\,dt$

$= \frac{2}{3}+\sqrt{t} - 2\sqrt{t} + c = \frac{2}{3}(x-2)\sqrt{x+1} + c$

iii) $\int x\cos x\,dx =$ (Integration by parts (IBP))

$= x\,\text{sen}x + \int \text{sen}x\,dx$

$= x\,\text{sen}x - \cos x + c$

iv) $\int e^x \sin x\,dx =$ (IBP)

$= e^x\sin x - \int e^x\cos x\,dx + c$ (IBP)

$= e^x\sin x - e^x\cos x - \int e^x\sin x\,dx +$

Then $2\int e^x\sin x\,dx = e^x\sin x - e^x\cos x$

$\Rightarrow \int e^x\sin x\,dx = \frac{1}{2}e^x(\sin x - \cos x)$

b) $\int \cos(x)\cos^{n-1}(x)\,dx =$ (IBP)

$= \sin(x)\cos^{n-1}(x) + \int \sin^2(x)\cdot(n-1)\cos^{n-2}(x)\,dx$

$= \sin(x)\cos^{n-1}(x) + \int (1-\cos^2(x))(n-1)\cos^{n-2}(x)\,dx$

$= \sin(x)\cos^{n-1}(x) + (n-1)\int \cos^{n-2}(x)\,dx - (n-1)\int \cos^n(x)\,dx$

Then $n\int \cos^n(x)\,dx = \sin(x)\cos^{n-1}(x) + (n-1)\int \cos^{n-2}(x)\,dx$

$\Rightarrow \int \cos^n(x)\,dx = \frac{1}{n}\sin(x)\cos^{n-1}(x) + \left(\frac{n-1}{n}\right)\int \cos^{n-2}(x)\,dx$

Exercise 3

a) $\int_{1/2}^{\infty} e^{-2x+1}\,dx = \lim_{M\to+\infty} \int_{1/2}^{M} e^{-2x+1}\,dx$

$= \lim_{M\to+\infty} \int_{0}^{2M-1} \frac{1}{2} e^{-t}\,dt = \lim_{M\to\infty} -\frac{1}{2} e^{-t}\Big|_{0}^{2M-1} =$

$= \lim_{M\to+\infty} \frac{-1}{2} e^{2M-1} + \frac{1}{2} = \frac{1}{2}$

b) $\int_{0}^{\infty} \frac{2x}{x^2+1}\,dx = \lim_{M\to+\infty} \int_{0}^{M} \frac{2x}{x^2+1}\,dx$ $\left(\text{substitution } \begin{array}{l} x^2+1 = t \\ dt = 2x\,dx \end{array}\right)$

$= \lim_{M\to+\infty} \int_{1}^{M^2+1} \frac{1}{t}\,dt = \lim_{M\to+\infty} \log(M^2+1) - 0 = +\infty$

c) $\int_{1/\pi}^{\infty} \sin(1/x)\,dx = \lim_{M\to\infty} \int_{1/\pi}^{M} \sin(1/x)\,dx$

Notice that $\lim_{x\to+\infty} \frac{\sin(1/x)}{1/x} = 1$

Hence $\exists\ N>0$ s.t. $\sin(1/x) \geq \frac{1}{2}\cdot\frac{1}{x}$ $\forall\, x > N$

Since $\sin(1/x) \geq 0$ for $x \geq \frac{1}{\pi}$ we have

$\lim_{M\to\infty} \int_{1/\pi}^{M} \sin(1/x)\,dx \geq \lim_{M\to\infty} \int_{N}^{M} \sin(1/x)\,dx$

$\geq \lim_{M\to\infty} \int_{N}^{M} \frac{1}{2}\frac{1}{x}\,dx$

$= \lim_{M\to+\infty} \frac{1}{2}\log(M) - \frac{1}{2}\log(N) = +\infty$

Exercise 4

a) We want to study the convergence of $\sum_{k=2}^{\infty} \frac{1}{k(\log k)^{\alpha}}$ (notice that the denominator would vanish for $k=0,1$).

Let us consider $f(x) = \frac{1}{x(\log x)^{\alpha}}$, defined on $\mathbb{R}^+$.

f is positive and decreasing, hence

$$\sum_{k=2}^{\infty} \frac{1}{k(\log k)^{\alpha}} = \sum_{k=2}^{\infty} f(k) \text{ converges } \iff$$

$$\int_2^{\infty} f(x)\,dx < +\infty.$$

Let us check:

$$\int_2^{\infty} f(x)\,dx = \int_2^{\infty} \frac{1}{x(\log x)^{\alpha}}\,dx \qquad \left(\text{substitution } \begin{array}{l}\log x = t\\ dt = \frac{1}{x}dx\end{array}\right)$$

$$= \int_{\log(2)}^{\infty} \frac{1}{t^{\alpha}}\,dt \qquad \text{that converges } \iff \alpha > 1.$$

b) As above let us define $f(x) = \frac{1}{(\log x)^{\log x}}$, on $\{x>1\}$

Since $\log(x)$ is positive and increasing on $\{x>1\}$, f is positive and decreasing. Then

$$\sum_{k=2}^{\infty} \frac{1}{(\log k)^{\log k}} = \sum_{k=2}^{\infty} f(k) \text{ converges } \iff$$

$$\int_2^{\infty} f(x)\,dx < +\infty.$$

Let us check:

$$\int_2^{\infty} f(x)\,dx = \int_2^{\infty} \frac{1}{(\log x)^{\log x}}\,dx \qquad \left(\text{substitution: } \begin{array}{l}\log x = t\\ x\,dt = dx\\ e^t dt = dx\end{array}\right)$$

$$= \int_{\log 2}^{\infty} \frac{e^t}{t^t}\,dt \le \int_{\log 2}^{+e^2} \frac{e^t}{t^t}\,dt + \int_{e^2}^{+\infty} \frac{e^t}{t^t}\,dt$$

$$\cancel{\le \int_{\log e^2}^{\infty} \frac{e^t}{t^t}\,dt \le \int_{e^2}^{\infty} \frac{e^t}{e^{2t}}\,dt = \int_{e^2}^{\infty}}$$

$\frac{e^t}{t^t}$ is bounded by some M on $[\log 2, e^2]$ since it is continuous

$$\le \int_{\log 2}^{e^2} M\,dt + \int_{e^2}^{+\infty} \frac{e^t}{e^{2t}}\,dt$$

$$\le e^2 \cdot M + \int_0^{+\infty} e^t\,dt \le e^2 \cdot M + 1 < +\infty$$

So the series is convergent